has taken the initiative to fill in the gaps. Their argument is, rather than betting the farm on a technology too futuristic for the currently available tools, why not give users the opportunity to ramp up as and when the time is right?

AMD's endeavour is only months away from your desktop. Later this year, the company will introduce ClawHammer, a 64-bit desktop processor that carries along a 32-bit legacy. This is the first in a series of CPUs, codenamed Hammer, that follow the path to 64-bit horizons. And, unlike Itanium, the Hammer series will carry the bridge that links their processors with older, 32-bit applications.

AMD's upcoming line up credits its foundations to a design called x86-64, giving it the ability to go both ways. This design works by adding several new instructions to the current x86 processor architecture. Doing this extends the chip architecture and it now becomes capable of addressing 64 bits of data. Currently, x86-based chips from both Intel and AMD address 32 bits of data. x86-64 allows AMD chips to support both.

The addition of 64-bit addressing allows chips to support larger amounts of physical memory which, among other things, speeds up large database programs typically used on servers. However, AMD's approach is directly opposite to that of Intel, which designed a completely new architecture for the Itanium called IA-64, to address 64 bits and run 32-bit applications where necessary. Intel is also said to be working on its own version of 64-bit extensions for x86. In practical terms, AMD is primarily concerned with getting 64-bit applications and operating systems for servers ported over to SledgeHammer, its server CPU. ClawHammers on the other hand will most likely use the regular 32-bit software—64-bit applications won't come to desktops until around 2004.

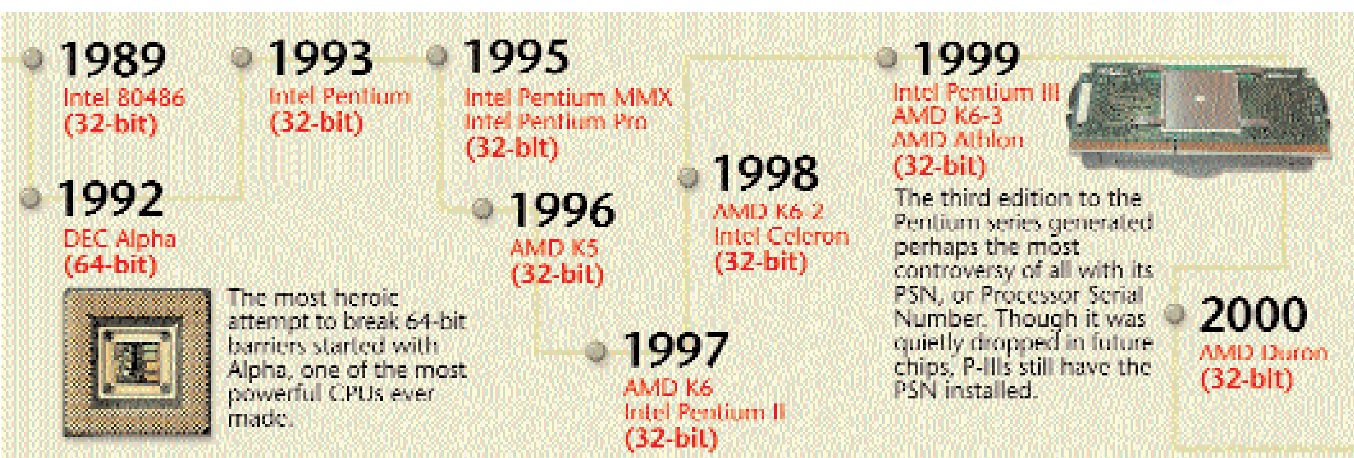
ClawHammer fills in the gaps that Itanium created between today's 32-bit apps and tomorrow's 64-bit technology

Moving upwards

Making the move to a 64-bit platform sounds simpler than it actually is. But getting down to the brass tacks, the entire shift requires a complete overhaul of your computer system, starting with the most obvious part, the CPU. The 64-bit chips of the future are totally new designs that simply won't jive with what's running today. They've been built anew from the ground up; that means everything dependent on the CPU needs an overhaul, too. This would include a CPU's chipset—a pair of logic chips used in guiding the data traffic throughout a computer system. And if the entire chipset needs to be changed, then your motherboard goes too.

On the software side, things are even more complex. Although 64-bit computing is definitely making some headway on the desktop front, this doesn't necessarily mean that software support is going to be swift, or for that matter, even reliable. One must remember that 64-bit is not new technology. For years since the Alpha chip, this new frontier of computing has had to endure scanty OS and application support. On one hand, chipmakers have been slapping price premiums on high-end chips. Knowing fully well that deep market penetration of these chips is a distant dream, OS makers such as Microsoft have stayed away from pouring millions of dollars on R&D for something that is not likely to generate significant returns. But with two of the largest chipmakers ready to place all their futuristic technology in the hands of the common computer user, this scenario is guaranteed to change.

Despite not having a 64-bit prototype on the market, let alone a fixed release date, it's comforting to see AMD's Hammer series garnering so much support from the software industry. Renowned OS makers such as SuSE, one of the top Linux developers and distributors, have already announced support for AMD's next generation architecture. SuSE has submitted updates for x86-64 to the Linux community with AMD holding on to the likely hope that these revisions will be a part of the upcoming Linux v2.6 kernel. The Linux kernel is built by top programmers including the OS's inventor,



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